Oliver Donaldson

Data Science Student | Data Engineering & Machine Learning

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PROFILE

Third-year BSc Data Science student at Victoria University of Wellington who builds data pipelines and the software around them in Python. Industry experience turning raw MongoDB and Kafka data into layered Snowflake reporting and automating an end-to-end operations workflow at a FinTech lender. Builds and runs Exoplanet Hunter, a live system that collates NASA and ESA archive data and ranks planet candidates for telescope follow-up; began its v3 rebuild as an installable Python library in October 2026.

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript, JavaScript, Java

Data Engineering: pandas, NumPy, ETL pipelines (Hevo, n8n), Snowflake, PostgreSQL, SQLite, MongoDB, Kafka, Parquet, Pandera data validation, DVC data versioning

Software & DevOps: Git and GitHub (branches, pull requests, code review, issues and project boards), GitHub Actions CI/CD, Docker, FastAPI (REST), pytest, Ruff, mypy, Prefect, Unix shell, Fly.io, Render

Machine Learning: TensorFlow / Keras, PyTorch, scikit-learn, 1D CNNs, MC-Dropout uncertainty, probability calibration, grouped k-fold cross-validation, MLflow, Optuna

Statistics & Visualisation: GLMs, shrinkage methods, ANOVA and design of experiments, clustering, A/B testing, R, matplotlib, Tableau, Grafana, Quarto

Governance & Ethics: AI governance, data privacy, regulatory frameworks

PROFESSIONAL EXPERIENCE

AI & Analytics Intern | *Harmoney, Financial Services (Platforms Team)* Nov 2025 – Feb 2026

- **End-to-end feature automation:** built the "Obligation Met Letter" loan-closure confirmation, replacing a manual Operations process handling ~70–80 requests per month. Implemented an 8-day eligibility buffer, system-event logging for audit trails, and a self-service download dashboard for customers.
- **Data engineering:** developed a multi-layer reporting framework in Snowflake, building SQL views that transformed raw MongoDB and Kafka data into Tableau-ready datasets for Finance teams.
- **Real-time monitoring:** built "StatsBot" using n8n, Kafka and MongoDB aggregation pipelines to serve real-time API ground-truth statistics, replacing legacy reporting systems.
- **AI governance:** authored GitHub Copilot system instructions to standardise PR reviews around security patterns and architectural consistency.
- **Stack:** Snowflake (SQL), MongoDB, Kafka, n8n, Tableau, TypeScript, React, Python, SendGrid.

Sailing Coach | *Royal Akarana Yacht Club* May 2019 – Jan 2024

- Led development programmes across a range of age groups, focusing on safety, confidence-building, skill development and leadership.
- Communicated complex technical skills to groups ranging from complete beginners through to competitive racing squads.

Catering Assistant | *Compass Group NZ / Air New Zealand Lounge* Sep 2023 – Present

- Deliver premium hospitality service in a high-volume lounge environment while studying full-time, balancing a fixed weekend roster against academic deadlines.

PROJECTS

Exoplanet Hunter | *Transit-vetting system for NASA TESS, Kepler & K2 data* 2026 – Present

github.com/OliverDonaldson/exoplanet-hunter-v2 · live vetting console

Self-directed project that decides which dips in a star's light curve are likely planets rather than eclipsing binaries or instrument noise, and ranks unresolved candidates by calibrated probability for follow-up. v2 is live; v3 is under way.

- **Data pipeline:** pulls labels and orbits from the NASA Exoplanet Archive and ExoFOP, light curves from MAST and stellar parameters from Gaia DR3, then cleans, detrends and phase-folds them into versioned training data (DVC on Cloudflare R2): 5,826 labelled stars and 12,472 catalogue signals.
- **Data quality:** Pandera schema, coverage and leakage gates stop a run instead of warning. A scheduled weekly refresh (Prefect) pulls new labels, re-runs the gates and retrains, but never replaces the served model without a person's sign-off.
- **Serving:** a calibrated 5-fold 1D CNN ensemble with MC-Dropout uncertainty behind a FastAPI REST service (Docker on Fly.io) and a web vetting console; about 3,900 unresolved TESS candidates scored. Tests, linting and the console build run in GitHub Actions; every change goes through a written-up pull request tracked on a GitHub project board.
- **Self-audit (Sep 2026):** found that light curves were folded in days where the code assumed orbital phase, so the "whole-orbit" view was a one-day window, and that gradient-boosted trees on 13 vetting features beat the served network on TESS ROC-AUC (0.933 vs 0.910). Both are published in the project write-up and scheduled ahead of any retrain.

- **v3 (from Oct 2026):** an 11-stage rebuild as a tested, installable Python library that scores uploaded light curves, with a sealed test set fixed before any model is trained, classical baselines the neural model must beat, and a Journal of Open Source Software paper as the goal.
- **Stack:** Python, pandas, NumPy, TensorFlow/Keras, scikit-learn, lightkurve, Pandera, FastAPI, Docker, Prefect, DVC, MLflow, GitHub Actions.

Finding Patterns in Policing Data | *DATA 301 team project, NZ Police & Stats NZ data* Jul – Sep 2026

- Team of three studying how assault victimisation varies with age across New Zealand; the group's exploratory data analysis report scored 100/100.
- **Ingestion pipeline:** built the team's Python/pandas loader for eight annual NZ Police exports (1.8 million victims) and Stats NZ population estimates. Processed data is rebuilt from the untouched raw files on every run, so a clean checkout reproduces it exactly.
- **Data integrity:** found and documented source traps that produce plausible but wrong data, such as a mislabelled dashboard extract with no age field and Stats NZ age codes that double every count when read as integers; the loader raises an error instead of carrying on.
- **Collaboration:** set up the shared repository, project board and contribution guide, and a GitHub Actions workflow that renders the Quarto report from a clean checkout on every push.

Study Tracker | *Interactive macOS desktop widget*

- Desktop widget for tracking university study (course progress, daily tasks, a knowledge-tree SVG that grows as tasks are completed), built in React JSX on the Übersicht engine, with state persisted to JSON on every change.

F.R.I.D.A.Y. | *Finance, Resources, Information, Data, Assistance and Yield* CS50x final project, 2021

- Financial-guidance web app built as the Harvard CS50x capstone, with a voice assistant for hands-free navigation and a responsive layout for phone and desktop.

EDUCATION

BSc Data Science | *Victoria University of Wellington* 2024 – Expected 2027

- **Relevant coursework:** DATA 202 Data Management & Programming, DATA 305 Machine Learning Techniques, AIML 231 Techniques in Machine Learning, AIML 232 Techniques in AI, DATA 303 Statistics for Data Science, STAT 292 Applied Statistics, DATA 101 Introduction to Data Science, STAT 193 Statistics in Practice.
- **In progress (Trimester 2, 2026):** DATA 301 Data Science in Practice, STAT 293 Applied Statistical Modelling, GEOG 215 Introduction to GIS & Science.

Law Coursework | *Victoria University of Wellington* 2024 – 2025

- Completed LAWS 121 Introduction to the NZ Legal System, LAWS 122 Introduction to Case Law, LAWS 123 Introduction to Statute Law and LAWS 297 Introduction to Legal Skills as part of a BSc/LLB conjoint, before specialising in Data Science.

NCEA Level 3 (Excellence) | *St Peter's College, Epsom* 2017 – 2023

- University Entrance; Level 3 and Level 2 Excellence endorsements; College Communications Prefect (2023), leading the prefect team's internal communications and starting the St Peter's College Speak podcast.

LEADERSHIP & ACTIVITIES

General Committee Member | *VicAI, Victoria Artificial Intelligence Society, VUW* Jun 2026 – Present

- Organising a student Project Expo: lightning talks from postgraduate students across data science, AI and cyber security, aimed at encouraging undergraduates to build their own portfolio projects.
- Ran volunteer rostering and member communications for the club stand at the university Clubs Expo.

Class Representative | *LAWS 297 and STAT 292, Victoria University of Wellington* 2025

- Represented two classes to teaching staff; recognised by the Students' Association.

Co-founder, SailSaver | *Young Enterprise Scheme* 2023

- Turned old sails into tote bags and donated 25% of profits; won Excellence in Business Management for central Auckland at the YES Tāmaki Makaurau Regional Awards.

Competing Athlete | *NZIFBB* 2025

- Competed in natural bodybuilding, requiring sustained discipline and long-horizon goal-setting alongside full-time study.

Youth Training Programme | *Royal New Zealand Yacht Squadron* 2022

- Selected for the Mastercard Youth Training Programme; represented the Squadron at the Harken Youth International Match Racing Cup, reaching the quarterfinals and finishing 6th.

CERTIFICATIONS

- **CS50x: Introduction to Computer Science**, Harvard University, 2021.
- **Certificate of Proficiency (Young Scholars Programme)**, University of Auckland, 2023: PHIL 105 Critical Thinking (B+) while at school.
- **Sail Coach Certification**, Yachting New Zealand, 2017.